

28/09/26

Sensitivity Analysis

the study of changes in optimal sol^{ns} due to the changes of LPP parameters (b, c, a_{ij}) are post-optimality Analysis (Sensitivity Analysis)

$$\left. \begin{array}{l} \text{Max } Z = cx \\ \text{Subject to } Ax = b \\ x \geq 0. \end{array} \right\} \text{LPP.}$$

- (i) variation in the cost vector c . ✓
- (ii) variation in the requirement vector b
- (iii) " " " Co-efficient matrix $A = [a_{ij}]$
↳ element a_{ij}
- (iv) Add/deletion of a new variable
- (v) Add/deletion of a new constraint.

$$\left. \begin{array}{l} \text{Max } Z = 12x_1 + 16x_2 \\ \text{Subject } 10x_1 + 20x_2 \leq 120 \\ 8x_1 + 8x_2 \leq 80 \\ x_1, x_2 \geq 0. \end{array} \right\} \text{LPP.}$$

$$\rightarrow c = (12, 16)$$

$$b = \begin{bmatrix} 120 \\ 80 \end{bmatrix}$$

$$A = \begin{bmatrix} 10 & 20 \\ 8 & 8 \end{bmatrix}$$

Final iteration

CB _j	C _j B.V	12	16 + ΔC _k	0	0	Soln
		x ₁	x ₂	s ₁	s ₂	
16 + ΔC _k	x ₂	0	1	1/10	-1/8	2
12	x ₁	1	0	-1/10	1/4	8
	Z _j	12	16	2/5	1	128
	Z _j - C _j	0	0	2/5	1	

$$C = (12, 16, 0, 0)$$

↑ ↑ ↑ ↑
x₁ x₂ s₁ s₂

$$Z_K = 16 \cdot 1 + 12 \cdot 0$$

$$Z_K^* = (16 + \Delta C_K) \cdot 1 + 12 \cdot 0$$

$$= 16 \cdot 1 + 12 \cdot 0 + \Delta C_K \cdot 1$$

$$= Z_K + \Delta C_K \cdot 1$$

$$Z_j - C_j \geq 0$$

Solution:

$$x_1 = 8, x_2 = 2$$

$$Z = 128$$

Three possibility after parameter changes

- The optimal soln is completely unchanged
- Same variables remain basic but their value changes.
- The basic soln changes entirely.

(i) variation of cost vector c

$$\left. \begin{array}{l} \text{Max } Z = cx \\ \text{Subject } Ax = b, x \geq 0 \end{array} \right\}$$

A = Co-efficient matrix

c = cost vector

b = requirement vector

$$\boxed{x_B = B^{-1}b} \rightarrow \text{solution.}$$

optimality condition: $z_j - c_j \geq 0$

$$\begin{aligned} z_j - c_j &= c_B B^{-1} a_j - c_j \\ &= c_B y_j - c_j \geq 0 \end{aligned} \quad \text{where } y_j = B^{-1} a_j$$

Subcase-I x_k is a non-basic variable.

$$c_k^* = c_k + \Delta c_k.$$

$\boxed{x_B = B^{-1}b}$ $\rightarrow$ solⁿ will be unchanged.

But $z_j^* - c_j^* \geq 0$ ($j=k$) for optimality

$$= c_B y_{jk} - (c_k + \Delta c_k) \geq 0$$

$$= z_k - c_k - \Delta c_k \geq 0$$

$$\Rightarrow \boxed{\Delta c_k \leq z_k - c_k}$$

Subcase-II x_k is Basic,

$$\begin{aligned} z_j^* - c_j &= c_B^* y_j - c_j \\ &= (z_j - c_j) - \Delta c_k y_{kj} \end{aligned}$$

$$\boxed{x_B = B^{-1}b} \rightarrow$$

optimality condition.

$$z_j - c_j \geq 0$$

$$\Rightarrow (z_j - c_j) - \Delta C_k y_{kj} \geq 0$$

$$\left. \begin{array}{l} \text{If } y_{kj} > 0 \Rightarrow \Delta C_k \geq - \frac{z_j - c_j}{y_{kj}} \\ \text{If } y_{kj} < 0 \Rightarrow \Delta C_k \leq - \frac{z_j - c_j}{y_{kj}} \end{array} \right\}$$

$$\max_{y_{kj} > 0} \left\{ - \frac{z_j - c_j}{y_{kj}} \right\} \leq \Delta C_k \leq \min_{y_{kj} < 0} \left\{ - \frac{z_j - c_j}{y_{kj}} \right\}$$